

The Tier-3 ASTRA MLP emulator — design, floor investigation, and inference prototype

ASTRA clustering project

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Abstract

This note specifies the first MLP emulator built on the Tier-3 pilot (`data/fullbox_tier3/`: 10 cosmologies $\times$ 50 HODs, all complete). It fixes the emulator *inputs*, the *target* data vector (prioritising the void/knot environments from the randoms, then the data, in monopole and quadrupole), the *cached dataset* that decouples slow disk reads from fast retraining, the *train/validation/test split* (leave-one-cosmology-out, not a random split), the *architecture*, and the *diagnostic plots*. It is the companion to `tier3_emulator_note.tex` (which sets the campaign strategy); this note is the operational recipe the code implements.

1 Inputs (20-D)

The emulator maps $\theta = (\theta_{\text{cosmo}}, \theta_{\text{HOD}}) \mapsto \xi(s)$. The Tier-3 block `c130–c181` varies the broader $w_0 w_a$ CDM+ set, so the cosmology input is **8 varying parameters** (read from `data/abacus_cosmologies_params.`

$$\omega_b, \omega_{\text{cdm}}, h, n_s, \alpha_s, N_{\text{ur}}, w_0, w_a.$$

(A_s and ω_{ncdm} are fixed across the block; σ_8 is a derived quantity and is kept as a label, not an input, to avoid collinearity.) The HOD input is the **12 yuan23 parameters** already used throughout the project: `LOGM_CUT`, `LOGM1`, `SIGMA`, `ALPHA`, `KAPPA`, `ALPHA_C`, `ALPHA_S`, `S`, `ACENT`, `ASAT`, `BCENT`, `BSAT`. Total input dimension **20**. All inputs are standardised (zero mean, unit variance over the training set).

2 Targets

The science priority is the **extreme environments** — voids (Q1) and knots (Q4) — **mostly from the randoms**, then from the data, in both multipoles. The primary target legs are therefore

$$\{\text{rand_q1}, \text{rand_q4}, \text{cross_full_rand_q1}, \text{cross_full_rand_q4}\}$$

followed by the data twins `{data_q1, data_q4, cross_full_data_q1, cross_full_data_q4}`, each in $\ell = 0$ and $\ell = 2$. The current pilot multipoles are stored on the coarse 15-bin $0\text{--}150 h^{-1}$ Mpc grid ($\sim 10 h^{-1}$ Mpc bins); the fine small-scale binning the strategy note calls for would require a pipeline re-run and is deferred. We train on the bins on disk and report accuracy as a function of a scale cut s_{cut} so the small-scale ($s < 40 h^{-1}$ Mpc) payoff region is read off directly.

3 Cached dataset — the key to cheap retraining

We will retrain many times (different target subsets, architectures, PCA truncations, loss weights). Re-globbing ~ 1000 run directories each time is wasteful, so a one-shot builder (`scripts/build_emulator_dataset.py`) writes a single compact archive `data/emulator_tier3/dataset.npz` holding the *full* stem set — all $17 \text{ stems} \times 2 \text{ multipoles} \times 15 \text{ bins}$ (510-D) — even though the first model trains on the void/knot subset. Arrays:

- `X_cosmo` ($N, 8$), `X_hod` ($N, 12$) — standardisable inputs;
- `Y` ($N, 510$) — concatenated $[\xi_0, \xi_2]$ over stems;
- `Ynoise` ($N, 510$) — per-bin ASTRA-iteration scatter (`xi*_std`);
- `s` (15,), `stem_labels`, `ell_labels`, `bin_index` — block bookkeeping;
- `cosmo_id`, `hod_id`, `sigma8` — per-row metadata for grouping/plots.

Target selection at train time is a column mask over `Y`; no disk re-read. The same builder writes `dataset_anchor.npz` from the 450 Fisher-grid runs (`data/fullbox/`, cosmologies `c000, c100--c105, c112, c113`), which are *not* in the training block and serve as an independent generalisation anchor in the LCDM neighbourhood.

4 Train / validation / test split

The decisive structural fact: there are only **10 cosmologies** but 500 runs. A random split over runs would place HODs of the same cosmology in both train and test, leaking the (sparse, hard) cosmology axis and giving an optimistic score. In deployment the emulator predicts at *new* cosmologies, so the split is made **along the cosmology axis**:

- **Headline: 10-fold leave-one-cosmology-out (LOCO).** Each fold trains on 9 cosmologies (450 runs) and tests on the held-out one (50 runs); rotating over folds tests every cosmology. This is the go/no-go metric.
- **Inner validation:** within each fold’s 9 training cosmologies, $\sim 15\%$ of the HODs (drawn across all 9, fixed seed) are held out for early-stopping and hyperparameter choices.
- **Per-fold fractions** $\approx 76\%$ train / 14% validation / 10% test, partitioned by cosmology — *not* a random 70/15/15 over runs.
- **External anchor:** the 450 Fisher-grid runs, never trained on, give a second test set in the dense LCDM region we ultimately care about.

LOCO is chosen over a fixed permanent holdout because 10 cosmologies is too few to also sacrifice a standalone test block.

5 Architecture

SUNBIRD-style feed-forward MLP in `torch` (2.10, CUDA available):

- Input 20-D (standardised) $\rightarrow$ 3–4 hidden layers (256–512 units, SiLU) $\rightarrow$ output;

- **Output via PCA** (~ 10 – 20 components on the standardised target vector) for a compact, decorrelated regression target;
- **Noise-weighted MSE** loss, weight $1/\sigma_{\text{noise}}^2$ per bin, for the pilot. (The covariance-weighted loss that matches the inference metric is the eventual goal, but the covariance is the known weak point of Tier-3, so we start diagonal.)
- **Ensemble** of ~ 5 independently-seeded MLPs for the emulator uncertainty (used in the pull/calibration diagnostic);
- Adam, early stopping on the inner validation loss.

6 Diagnostic plots

Acceptance is judged per s -bin against two yardsticks already used at Tier-2: the **ASTRA noise floor** (mean per-iteration ξ_{std} , which no emulator should be expected to beat) and the **signal spread** (std of the training vectors, i.e. what there is to learn). A good emulator has LOCO RMS $\sim$ noise floor and $\ll$ spread, especially at $s < 40 h^{-1}$ Mpc. Plots:

1. **Learning curves** — train/validation loss vs epoch, per fold (overfit check).
2. **LOCO prediction vs truth** — $s^2\xi$ for void/knot, data & random, $\ell = 0, 2$, on the held-out cosmology.
3. **Error vs yardstick** — per-bin LOCO RMS against noise floor and signal spread vs s , zoomed to $s < 40 h^{-1}$ Mpc (the acceptance plot).
4. **Pull histograms** — $(\hat{\xi} - \xi)/\sigma_{\text{ens}}$ should be $\mathcal{N}(0, 1)$ (ensemble-uncertainty calibration).
5. **Accuracy vs scale cut** — RMS/noise integrated for $s < s_{\text{cut}}$, directly answering “sub-noise at $s < 40$ ”.
6. **Fractional-error heatmap** — (stem $\times \ell$) vs s -bin, colour = median |frac. error|; an at-a-glance map (expect random quadrupoles weakest, per prior findings).
7. **External-anchor prediction vs truth** — the same curves on the 9 Fisher cosmologies.

7 Caveats

Ten cosmologies over an 8-D cosmology input is thin: a poor LOCO result means “need the full c130–c181 $\times$ 120 campaign,” not “ASTRA fails.” The coarse 15-bin grid limits the very small-scale resolution that carries most of the Fisher signal. The HOD response is assumed smooth and the ASTRA-iteration noise (3 iterations in the pilot) sets a floor the emulator cannot cross. These are the questions the diagnostics are designed to expose.

8 Results I: the floor investigation

The pilot was run and the diagnostics above executed. The headline question “can a monopole+quadrupole emulator reach inference-grade accuracy?” decomposed, through a sequence of cheap tests, into a clear and ultimately encouraging picture. All “ $\times CV$ ” numbers are ratios of the held-out RMS to the cosmic-variance standard deviation at the $2 h^{-1}$ Mpc Gpc box volume (see the *CV correction* subsection below for an important denominator fix).

8.1 The pilot is extrapolation-limited, not broken

Leave-one-cosmology-out training gives physical predictions and 7/10 folds beat the mean, but the held-out error is large and *varies enormously by cosmology*: hull-centre cosmologies are predicted well, hull-edge ones badly. The learning curve versus the number of training cosmologies (Fig. 1) is essentially flat across the 2–9 cosmologies the pilot can probe — but those ten cosmologies are spread every fifth point over the w_0w_a CDM hull, so this test cannot probe the *density* the full campaign would add. The conclusion is that the pilot is **cosmology-extrapolation limited**: the cosmology axis is too sparsely sampled, not that the method fails.

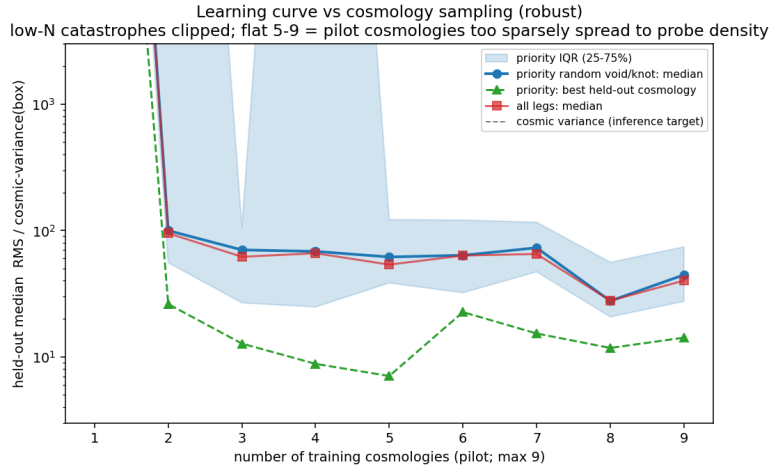


Figure 1: Held-out error versus number of training cosmologies (robust median over random subsets; RMS/spread panel is CV-estimate-noise-free). Flat across 2–9 cosmologies: at the pilot’s sparse spacing, adding points does not shrink the inter-cosmology gaps. The campaign changes the *density*, which this proxy cannot see.

8.2 Ruling out the suspects

A within-cosmology test (5-fold over each cosmology’s HODs, *zero* cosmology extrapolation) isolates a second, smaller floor (Fig. 2). We then eliminated its candidate causes one by one:

- **HOD sample size — ruled out.** The within-cosmology floor plateaus by ~ 30 training HODs and does not improve out to 70 (we ran c000 to 100 HODs to check; Fig. 3).
- **Model class / loss / representation — ruled out.** A bake-off of the MLP against a PCA+GP (covariance-weighted, amplitude-factored residual target) and a PCA+Ridge all tie (Fig. 3).
- **Iteration noise — ruled out** earlier (emulator error $\gg$ the ASTRA per-iteration scatter).
- **Binning — ruled out.** A zero-compute coarsening proxy (rebin the existing data to 3–15 bins) shows the modelling-quality metric (RMS/spread, immune to CV-estimate noise) is flat with resolution (Fig. 4); the apparent gain toward finer bins in RMS/CV is a noisy-denominator artifact. Finer binning is therefore *not* a strong lever.

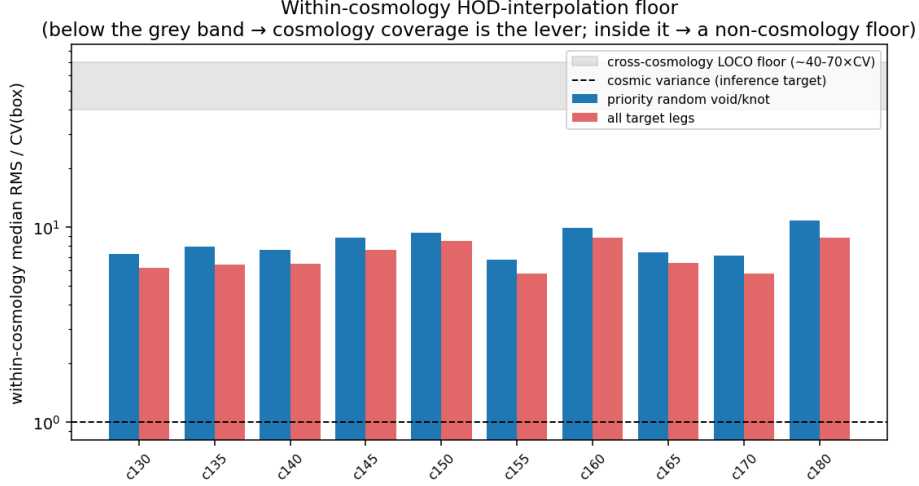


Figure 2: Within-cosmology HOD-interpolation floor per cosmology (zero cosmology extrapolation). A uniform residual floor remains even with no extrapolation; its cause is isolated in the following tests.

8.3 Resolution: the floor is monopole-vs-quadrupole

Splitting the within-cosmology floor by leg (Fig. 5) reveals the real structure: it is not data-vs-random but **monopole-vs-quadrupole**. The *monopoles* (both data and random) emulate well, several at or below cosmic variance; the *quadrupoles* carry the entire floor. This matches the independent iteration experiment (random quadrupoles are noise-limited at 3 iterations, improving toward 10): **the quadrupole floor is the 3-iteration ASTRA label noise on the velocity-dependent $\ell = 2$ signal**, not binning, model, or HOD count. The “ $\sim 6\text{--}8\times$ floor” quoted from aggregate metrics was great monopoles averaged with noisy quadrupoles.

8.4 A CV-denominator correction

While building the inference test we found a bug in the cosmic-variance helpers: they read the per-cosmology *mean* multipole (`xi0`, 9 samples) instead of the per-subbox array (`xi0_all`, 576 pooled samples), so “CV” was a cosmology-variation, not cosmic variance. After the fix every absolute $\times CV$ number shifts by $\sim 0.6\text{--}5\times$ per bin, but *all relative conclusions above hold* (they compare ratios with the same denominator on both sides), as do the RMS/spread results (which never use CV). The corrected absolute numbers are quoted in the next section.

9 Results II: most-informative scales and cosmology response

Two model-free diagnostics characterise where the cosmological information lives and how the legs respond to cosmology, using all 19 cosmologies on disk.

Scale. A cumulative-information analysis (Fig. 6) shows the **monopole** cosmological information is concentrated at **small scales** ($\sim 50\%$ by $s \approx 30\text{--}55 h^{-1} \text{ Mpc}$), while the **quadrupole** information sits at **large scales** ($\gtrsim 100 h^{-1} \text{ Mpc}$, the Kaiser regime). Small scales carry both the highest cosmology signal and the highest HOD nuisance — which is why the emulator is needed there.

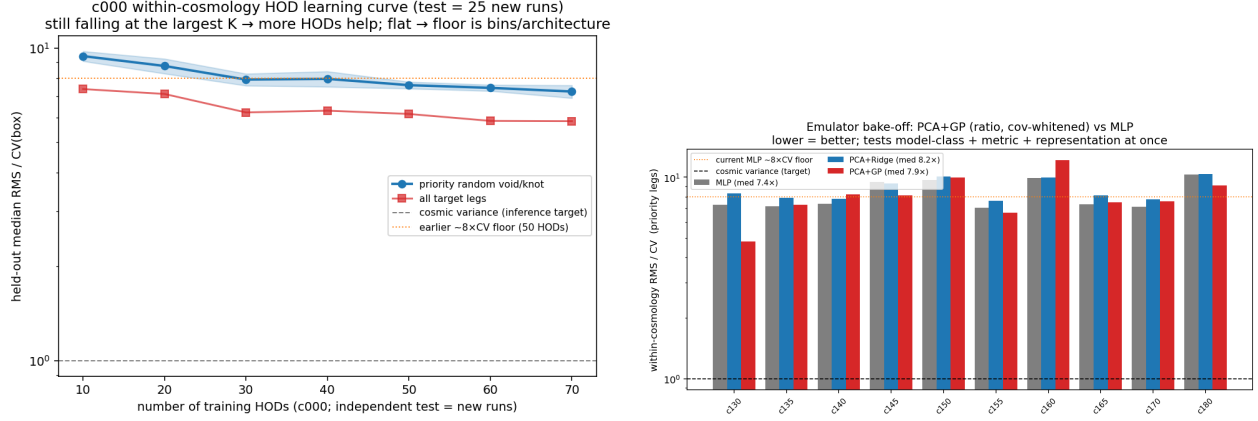


Figure 3: *Left*: c000 held-out error vs. number of training HODs (100 HODs available): plateaus by ~ 30 — HOD count is not the limiter. *Right*: model bake-off, within-cosmology floor per cosmology — MLP, PCA+Ridge and PCA+GP tie, so model/loss/representation is not the limiter.

Response. Single-parameter Fisher $\pm$ pairs (Fig. 7), HOD-marginalised and split by environment, give the clean attribution the broad Latin-hypercube cannot: ω_{cdm} is the largest, cleanest mover; σ_8 /amplitude is bias-degenerate (at fixed number density a *lower* σ_8 can give a *higher* ξ); and void and knot respond *differently* to each parameter — the degeneracy-breaking that motivates ASTRA, visible by eye.

10 Results III: monopole emulator and inference prototype

Guided by the floor result, we built the inference-ready piece: a **monopole** forward model and a SUNBIRD-style recovery test, both on the legs that are clean (the void *data* monopoles are excluded — near their ξ zero-crossing the CV vanishes and RMS/CV diverges).

Cross-cosmology accuracy (the gate). The corrected monopole leave-one-cosmology-out (Fig. 8) is a textbook interpolation-vs- extrapolation split: the nine Fisher cosmologies (dense LCDM neighbourhood) sit at $\sim 1.1 \times \text{CV}$ (c000 at $1.0 \times$, i.e. at cosmic variance), while the ten broad tier-3 cosmologies are extrapolation-limited at $\sim 18 \times \text{CV}$. So a monopole inference is genuinely inference-grade *near LCDM today*; the broad prior needs the campaign’s coverage.

Inference recovery. The likelihood uses the emulator $\mu(\theta)$, the subbox C_{CV} , and an emulator-error C_{emu} from the Fisher-set LOCO residuals, with **emcee** over the four LCDM parameters (HOD fixed = prototype). Two mocks (Fig. 9):

- **Synthetic** (μ at a displaced cosmology + a draw from $C_{\text{CV}} + C_{\text{emu}}$): $\chi^2/\text{dof} = 0.55$, all four parameters recovered within 1σ at the displaced truth — the sampler/likelihood/emulator-gradient **machinery is validated**.
- **Real c000** ($C_{\text{tot}} = C_{\text{CV}} + C_{\text{emu}} + C_{\text{label}}$): the calibration removes the gross bias of an earlier broken attempt (n_s from -30σ to -1.7σ), but $\chi^2/\text{dof} = 13.5$ and ω_b at $+3.2\sigma$ remain — residual per-run emulator bias beyond the averaged C_{emu} , plus the fundamental same-phase-mock limitation (no independent realisation is available under the no-new-HOD constraint).

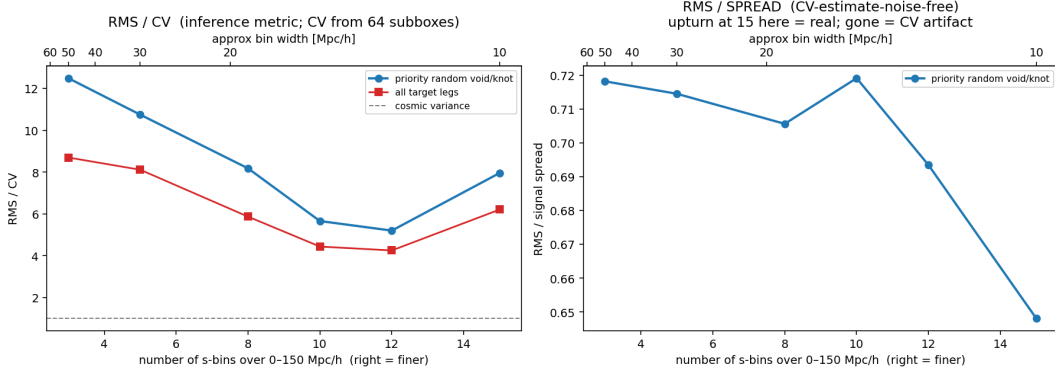


Figure 4: Coarsening-trend proxy. RMS/CV (left) is U-shaped, but the upturn at the finest bins vanishes in RMS/spread (right), which is flat — so the upturn was a CV-denominator artifact and finer binning is not a strong lever ($\lesssim 10\%$ over $50 \rightarrow 10 h^{-1}$ Mpc, far short of the order-of-magnitude needed).

11 Status and next steps

Summary. The emulator floor is understood and is not a wall: monopoles are inference-grade near LCDM ($\sim 1 \times \text{CV}$), the only true limiters are *cosmology coverage* (extrapolation, $\rightarrow$ the campaign) and *quadrupole iteration noise* ($\rightarrow$ a ≥ 10 -iteration run). The SUNBIRD-style inference pipeline is built and its machinery validated on synthetic mocks; a clean real recovery is gated by emulator realism and the absence of independent-phase mocks.

Minimal coverage run (in progress). Because coverage is the gate and HODs plateau by ~ 30 , the minimal step that fits the available allocation is **all 52 cosmologies $\times$ 20 maximin HODs $\times$ 3 iterations** (the 10 pilot cosmologies reused), ~ 840 new runs, ~ 490 node-hours on the regular queue (`select_tier3_min.py`, `queue/launch_tier3_minimal.sh`). The payoff test: does the broad-prior monopole LOCO drop from $\sim 18 \times$ toward $\sim 1 \times \text{CV}$ as the cosmology axis fills from 10 to 52?

Remaining levers. (1) the full **c130–c181 $\times$ 100 $\times$ 10** campaign (free on the overrun queue) for coverage *and* quadrupoles; (2) HOD marginalisation (currently fixed); (3) an external pre-populated mock suite for an honest covariance and genuinely independent recovery mocks.

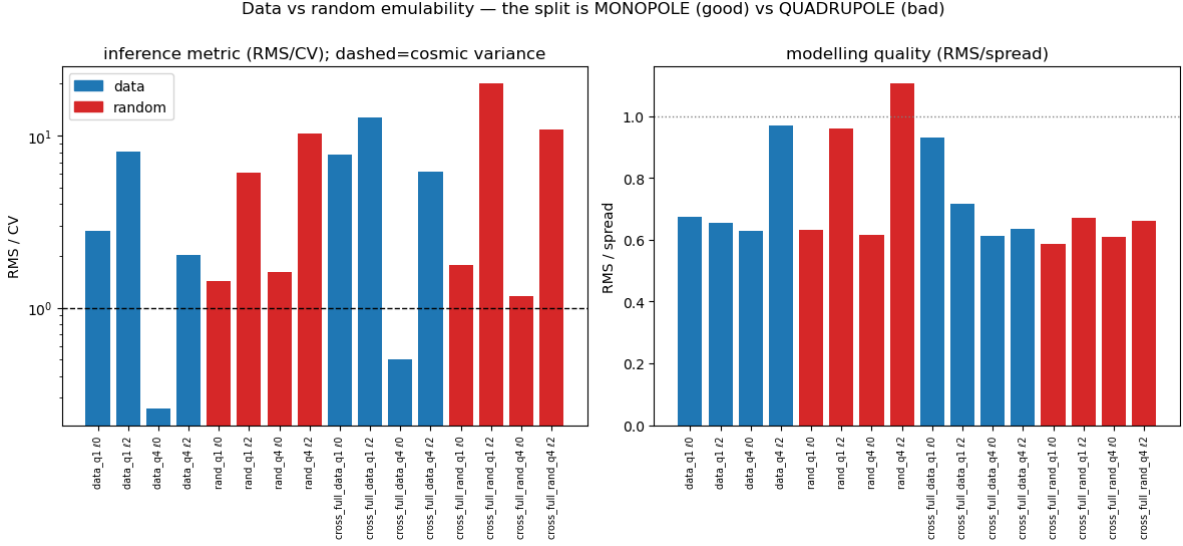


Figure 5: Per-leg within-cosmology emulability. The split is monopole (good) vs quadrupole (bad), in *both* data and random families. Monopoles are the inference-ready legs; the quadrupole floor is iteration-noise-limited.

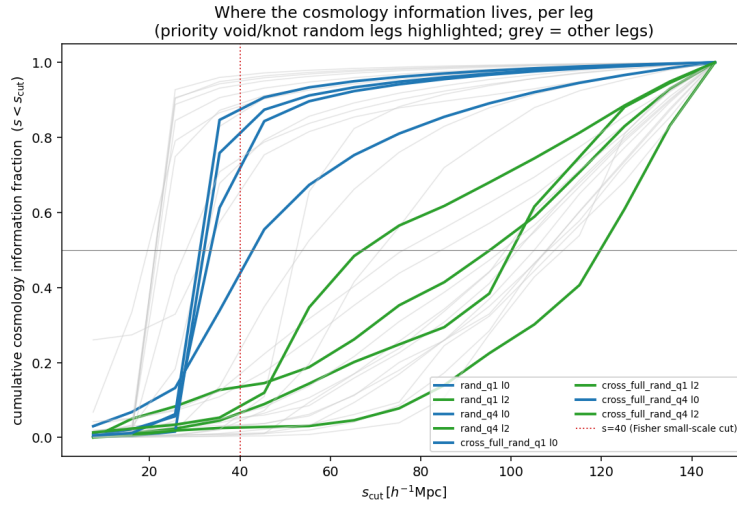


Figure 6: Cumulative cosmology-information fraction vs scale cut, per leg. Monopole ($\ell = 0$) curves saturate by $s \sim 50 h^{-1} \text{Mpc}$ (small-scale information); quadrupole ($\ell = 2$) curves rise slowly to large scales.

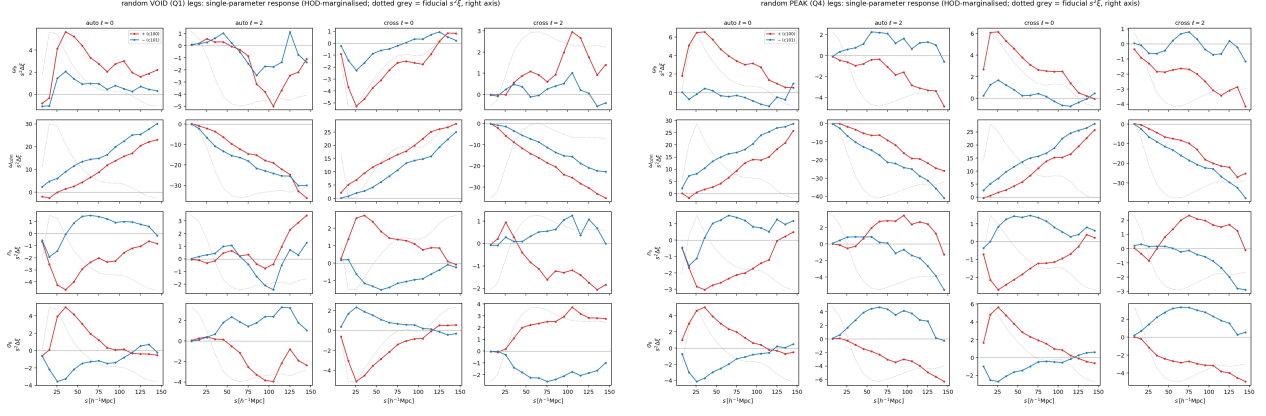


Figure 7: Single-parameter response of the random void (left) and knot (right) legs from the Fisher $\pm$ pairs (shift $s^2\Delta\xi$ for $+/-$; rows are the four LCDM parameters). The $+/-$ mirror symmetry indicates a clean linear response; ω_{cdm} dominates; void and knot differ.

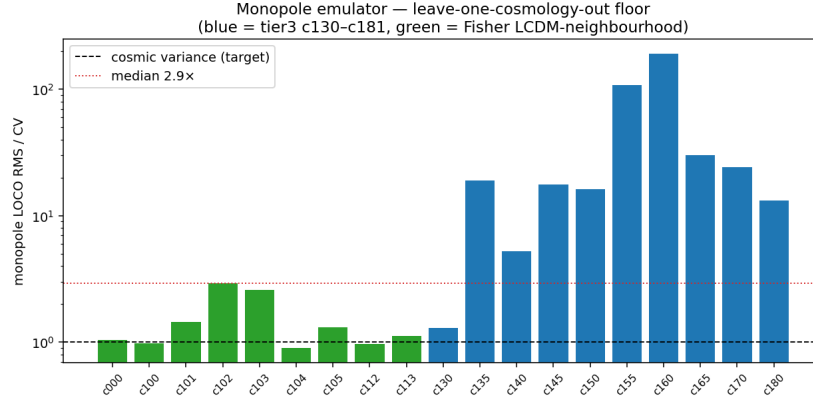


Figure 8: Monopole emulator leave-one-cosmology-out floor (corrected CV). Green = the nine Fisher LCDM-neighbourhood cosmologies ($\sim 1.1 \times \text{CV}$, interpolation); blue = the ten broad tier-3 cosmologies ($\sim 18 \times \text{CV}$, extrapolation). The gap is the cosmology-coverage gate.

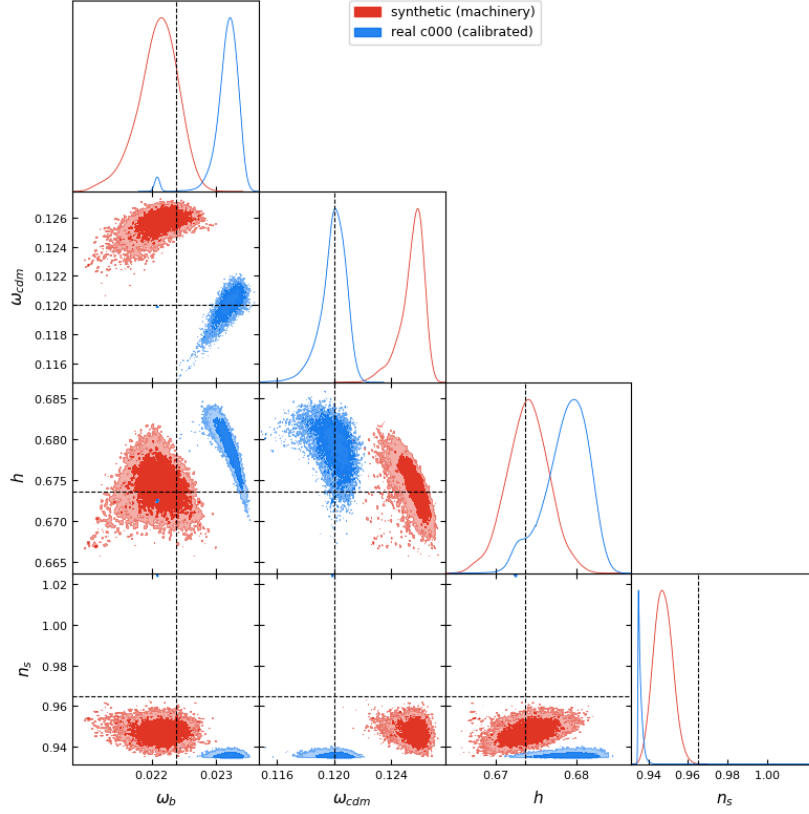


Figure 9: Recovery test (four LCDM parameters; dashed black = c000 truth). Red: synthetic mock — recovers the displaced injection within 1σ ($\chi^2/\text{dof} = 0.55$), validating the pipeline. Blue: real c000 mock with calibrated covariance — much improved but not yet clean ($\chi^2/\text{dof} = 13.5$), limited by per-run emulator realism.